

Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

- **HTTP – HyperText Transfer Protocol** – Used to transfer web pages between a browser and a web server.
- **HTTPS – HyperText Transfer Protocol Secure** – Secure version of HTTP that encrypts data during transmission.
- **FTP – File Transfer Protocol** – Used to transfer files between computers over a network.
- **SMTP – Simple Mail Transfer Protocol** – Used to send emails between mail servers and email clients.

What is the difference between IPv4 and IPv6? (2 lines)

- **IPv4** uses **32-bit addresses** and supports about **4.3 billion unique IP addresses**.
- **IPv6** uses **128-bit addresses**, providing a much larger number of unique IP addresses and improved security.

Run nslookup google.com on your system and write down the IP address you get.

Name: [google.com](https://www.google.com)

Addresses: 2404:6800:4000:100c::65

What do these HTTP status codes mean?

- **200** – OK (The request was successful.)
- **301** – Moved Permanently (The resource has been permanently moved to a new URL.)
- **404** – Not Found (The requested page or resource could not be found.)
- **500** – Internal Server Error (The server encountered an unexpected error.)
- **403** – Forbidden (Access to the requested resource is denied.)

Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

- **Root (.)** – The highest level of the DNS hierarchy.
- **TLD (Top-Level Domain)** – **.com**
- **Domain** – **google**
- **Subdomain** – **www**

Example: In www.google.com, **www** is the subdomain, **google** is the domain name, **.com** is the top-level domain, and all of them are managed under the **root (.)** DNS level.

